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Full Paper

Study of TATP: Influence of Reaction Conditions on Product Composition

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Received: June 15, 2008; revised version: January 7, 2009

DOI: 10.1002/prep.200800044

Abstract

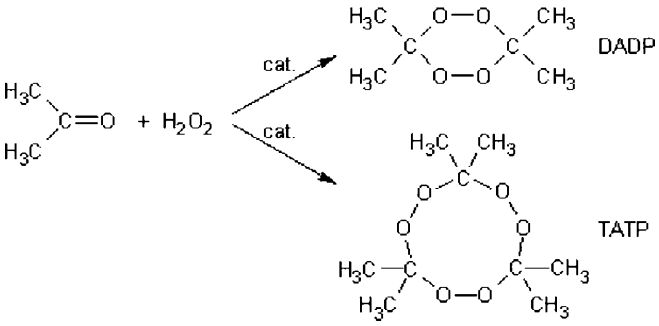
The influence of reaction conditions (temperature, catalyst type, catalyst concentration – represented as molar ratio catalyst/acetone n_c/n_a) on the composition of the product formed from the reaction of acetone and hydrogen peroxide (30%) under acid catalysis was studied. 3,3,6,6,9,9-Hexamethyl-1,2,4,5,7,8-hexoxonane (TATP) was found to be the major product when the content of the catalyst in the reaction mixture is low ($n_c/n_a \leq 0.5$). A single side product (peak 10.2) in an amount ranging from 1.5 to 8% of the total peak area was present in all the prepared samples. Three other side products were found when catalyzing by hydrochloric and nitric acids. Temperature and catalyst type did not have a significant influence on the composition of the product at low catalyst concentration. Increasing the catalyst concentration led to the formation of 3,3,6,6-tetramethyl-1,2,4,5-tetroxane (DADP) either as a co-product of TATP or as an exclusive product depending on the concentration of the catalyst.

Keywords: DADP, 3,3,6,6,9,9-Hexamethyl-1,2,4,5,7,8-hexoxonane, HPLC, TATP, 3,3,6,6-Tetramethyl-1,2,4,5-tetroxane

1 Introduction

The reaction of acetone with hydrogen peroxide gives in several reaction stages various types of organic peroxides of acetone. The cyclic peroxides (dimer and trimer) precipitate from the reaction mixture only in the presence of acid catalyst. The reaction mechanisms of their formation were investigated by several authors [1–5]. Depending on the reaction conditions (primarily pH, concentration of reagents, temperature, and environment), two cyclic organic peroxides are often reported to form – the dimer (3,3,6,6-tetramethyl-1,2,4,5-tetroxane (DADP)) and the trimer (3,3,6,6,9,9-hexamethyl-1,2,4,5,7,8-hexoxonane (TATP)) (see Scheme 1).

The reaction of acetone and hydrogen peroxide in the presence of acidic catalyst has been reported to yield either DADP or TATP. The reported reaction conditions were similar and it is unclear why some researchers obtained DADP [6–14] while others TATP [6, 11, 14–19].



Scheme 1. TATP and DADP formation from acetone and hydrogen peroxide in the presence of an acidic catalyst.

The objective of this work is to clarify the confusion regarding the influence of reaction conditions on the composition of the product obtained from the reaction of 30% hydrogen peroxide and acetone under acid catalysis, particularly with respect to DADP. The focus is aimed at the influence of the catalyst type, temperature during reaction, the catalyst to acetone molar ratio (n_c/n_a), and the order of reagents' mixing.

2 Experimental Part

Warning – cyclic acetone peroxides have the character of primary explosives and hence are sensitive to impact, friction, electric discharge, and flame. The synthesis and handling of TATP and other organic peroxides presented in this paper are dangerous operations that require standard safety precautions for handling the primary explosives!

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2.1 Sample Preparation

The DADP standard for HPLC was prepared according to both the procedures of Dubnikova et al. [11] and was recrystallized from methanol.

All raw TATP samples were prepared by the following procedure: 2.5 mL (34.1 mmol) of acetone (grade purum) was mixed with 4.35 mL (42.6 mmol) of 30% hydrogen peroxide (grade purum). The molar ratio of hydrogen peroxide to acetone was, therefore, always 1.25. The catalyst was then added to the stirred reaction mixture. The following catalysts (grade p.a. or purum) were used: HCl (37%), H_2SO_4 (31% and when studying influence of the catalyst concentration – Section 3.1.2 – also 65 and 96%), HNO_3 (65%), $HClO_4$ (70%), CH_3SO_3H (100%), HCOOH (100%), CH_3COOH (98%), and $SnCl_4 \cdot 5H_2O$ (solid). The amount of the catalysts used in the preparations is listed in Table 1. It is expressed as a molar ratio of catalyst to acetone and abbreviated as n_c/n_a. The temperature was controlled according to one of the following two temperature programs (A, B):

A The reaction temperature in majority of the experiments (group A) was controlled during the course of reagents addition and further maintained for 1 h. After this time the reaction mixture was allowed to acquire room temperature at which it was kept without stirring or cooling for 24 h. In cases when TATP did not form or was formed in low yield within the first 24 h, additional 6 days were allowed for the reaction at room temperature. These experiments were carried out at 0, 10, 20, 30, and 40 °C applying cooling by ice bath or heating by water bath as necessary.

B In the second set of experiments (group B) the temperature was held constant not only during reagent mixing but also for a period of 24 h following the mixing. In cases when TATP did not form or was formed in a low yield, the reaction mixture was kept at the same temperature for additional 13 days. These experiments were carried out at – 20 and 0 °C. The reaction was cooled using either a dry-ice/acetone bath or an ice bath and the reaction mixture was stored in a standard freezer or refrigerator.

The resultant precipitate was in all cases collected by filtration, washed twice with distilled water, once with 1% solution of sodium carbonate, then again with water (until neutral pH by indicator paper) several times, and finally dried at room temperature. These products were not purified by recrystallization before the HPLC analysis. All samples were prepared and measured in duplicate or triplicate.

A small number of samples were prepared by adding acetone to the mixture of hydrogen peroxide with catalyst and peroxide to the mixture of catalyst with acetone. These samples were used for the determination of the influence of the order of mixing (Section 3.4).

Table 1. Summary of the reaction conditions used for the preparation of cyclic acetone peroxide samples: amount of the catalysts and appropriate temperatures (program A unless otherwise specified).

Molar Ratio n_c/n_a	0.025	0.05	0.1	0.25	0.5	1.0	2.5
Temperature (°C)	20	20	20	– 20[a], 0[a], 0,10, 20, 30	20	20	20

[a] Temperature program B.

2.2 Sample Analysis

The liquid chromatographic system consisted of the following components (all from Agilent 1200 series): degasser G1322A, pump G1311A, autosampler G1329, column thermostat G1316A, and diode array detector G1315D operated by software HP Chem Station B03.01. Isocratic separation using solvent mixture consisting of 60% acetonitrile and 40% water with a flow rate of 1 mL min^{-1} was used. The injection volume was 10 μL, column material was BDS Hypersil C-18, particle size 5 μm, and pore size 12 nm, column dimensions 250 mm × 4.6 mm, temperature 24 °C. The chromatograms were evaluated from the absorbance at 210 nm.

The samples were measured within 24 h of preparation, generally on the next day after drying overnight at room temperature. The solutions for HPLC were prepared by adding ~8 mL of mobile phase (acetonitrile/water 60:40) to 30 – 80 mg of product in a volumetric flask and dissolving it with the aid of sonication before diluting it to 10 mL.

2.3 Sample Disposal after Analysis

Excess samples of TATP and DADP after analysis were destroyed by burning them in small portions (less than 1 g) in an unconfined thin layer in a fume hood. According to the author's experience with hundreds of samples, this may be considered a relatively safe procedure (similar to burning nitrocellulose). It is recommended to wear ear and eye protection and to keep the fume hood partially open. This method of peroxide destruction is, however, not recommended for samples of larger quantities. Combustion of larger unconfined samples (70 g) on our test polygon had been observed to undergo deflagration to detonation transition and would lead to serious damage if carried out in fume hood!

3 Results and Discussion

The DADP standards prepared according to both the procedures of Dubnikova et al. [11] were practically identical. Preparation of pure TATP was not successful as it was always accompanied by a small amount of a side product (peak 10.2) that showed in the chromatogram as a small peak eluting at longer retention time (RT) (Figure 1). This substance is reported to be tetraacetone tetraperoxide; 3,3,6,6,9,9,12,12-octamethyl-1,2,4,5,7,8,10,11-octaoxacyclododecane (TeATeP) by Schulte-Ladbeck et al. [20], Pena Quevedo et al. [21] or as a

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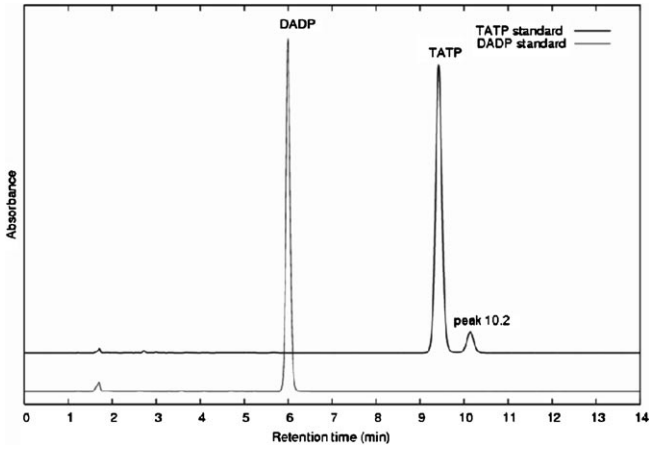


Figure 1. Chromatograms of standards of DADP prepared by procedure of Dubnikova et al. [11] and TATP prepared using hydrochloric acid ($n_c/n_a = 0.25$). The chromatogram of TATP standard shows a small amount of an unknown substance (peak 10.2) always present in TATP samples.

structural conformer of TATP by Widmer et al. [22]. We did not analyze this side product further. It was impossible to remove this substance by repeated recrystallization or sublimation. As we have found and published previously [23, 24] it is possible to reduce the amount of this substance by long-term thermal treatment at higher temperatures.

3.1 Influence of the Catalyst Type

The reaction of acetone with hydrogen peroxide has been found to yield products of varying compositions and stabilities depending on the reaction conditions of the synthesis. The most important factors were found to be the type and the concentration of the catalyzing acid in the reaction mixture. Overall concentration is practically impossible to be determined due to a change in volume of the reaction mixture and the formation of the solid product. We have, therefore, used the molar ratio of catalyst to acetone (n_c/n_a) before mixing instead of the concentration.

3.1.1 Influence of Catalyst to Acetone Ratio

The samples prepared using various catalysts and molar ratios are summarized in Table 2. At low concentrations of the catalyst, the products contained only TATP and small amount of the side product (peak 10.2) ($n_c/n_a \leq 0.5$). None of the samples contained DADP. Bellamy [25] reported that the purity of TATP depends on the type of catalyst and that use of sulfuric acid leads to TATP contaminated by DADP. The concentration of the sulfuric acid was not reported and is, therefore, difficult to compare to our results.

The ratio of areas of chromatographic peaks TATP/peak 10.2 did not depend significantly on the type of catalyst and molar ratio n_c/n_a (from 0.025 to 0.25). A decreasing amount of the side product (peak 10.2) in the product was observed in samples with $n_c/n_a \geq 0.5$.

The formation of DADP was observed for samples with high molar ratios $n_c/n_a \geq 1$. The course of the reaction depended on the type and concentration of the catalyst (see Table 3). The formation of two new compounds was observed in the product from hydrochloric acid and will be discussed later. DADP forms only when strong acid is used as catalyst and does not form when acetic and formic acids are used. When concentrated inorganic acids are used as catalysts in high molar ratios n_c/n_a, decomposition of the

Table 2. Influence of catalyst to acetone ratio on the composition of prepared samples expressed as the ratio of chromatographic peak areas of TATP and peak 10.2 (at 20 °C).

Catalyst	Molar Ratio n_c/n_a						
	0.025	0.05	0.1	0.25[a]	0.5	1	2.5
HCl	11.5	12.4	12.4	15.5	94.0	20.9[b]	–[b),c]
H_2SO_4 (31%)	11.2	12.4	13.0	15.3	58.7	32.7	84.3
HNO_3	13.4	13.0	20.9	15.6	60.2	57.8[d]	DADP[e]
$HClO_4$	13.7	14.1	18.7	25.4	48.4	DADP[e]	DADP[e),f]
CH_3SO_3H	12.2	11.6	11.4	20.1	14.4	60.4	DADP[c]
$SnCl_4$	12.9	12.7	14.8	18.5	13.3	29.8[g]	–
HCOOH	24.8	24.4	22.4	18.6	18.3	19.0	19.7
CH_3COOH	21.2	22.0	22.3	20.2	23.8	19.5	18.3

[a] Average values from repeated experiments.
[b] Sample contains a new substance eluting at 13.0 min.
[c] Sample contains only a small amount of TATP, the side product (peak 10.2) is either missing or hardly identifiable.
[d] Sample also contains DADP.
[e] Sample contains only DADP.
[f] The reaction mixture decomposed three times during the addition of the catalyst. The fourth mixture survived the addition and was analyzed.
[g] The mixture decomposed violently after complete addition of the catalyst. The result reported is from the analysis of the undecomposed residue. Based on this experience, it was decided not to carry out experiments at ratio $n_c/n_a = 2.5$.

Table 3. Influence of higher molar ratio n_c/n_a on the composition of products (reaction temperature 20 °C, all peaks were used to determine the 100% area).

Catalyst	Peak Area (%)					
	$n_c/n_a = 1$			$n_c/n_a = 2.5$		
	DADP	TATP	Peak 10.2	DADP	TATP	Peak 10.2
HCl[a]	Traces	90.1	4.3	Traces	3.0	Traces
H_2SO_4 (31%)	–	96.8	3.0	–	97.5	1.2
HNO_3[b]	Traces	82.3	1.4	71.1	Traces	–
$HClO_4$	99.1	Traces	–	98.9	–	–
CH_3SO_3H	–	97.2	1.6	89.9	10.1	–
$SnCl_4$[c]	–	96.0	3.2	Not Tested		
HCOOH	–	94.5	5.0	–	94.8	4.8
CH_3COOH	–	94.6	4.8	–	94.2	

[a] Products contained another side product (see Section 3.1.4).
[b] Products contained another side product (see Section 3.1.3).
[c] The mixture decomposed violently. The results in the table refer to the solid residue remaining after the decomposition.

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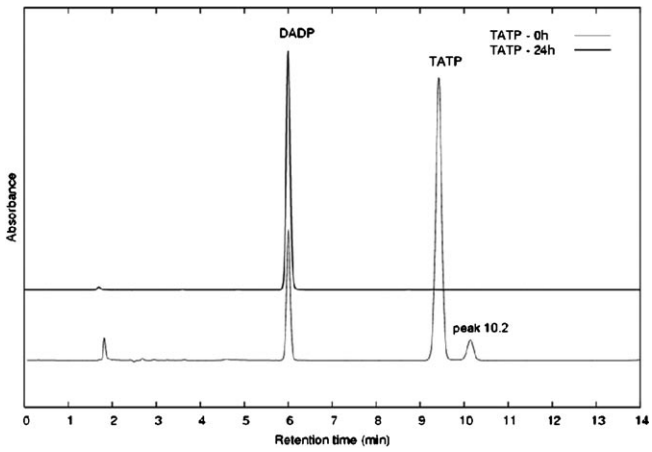


Figure 2. Chromatogram of TATP prepared from perchloric acid ($n_c/n_a = 2.5$) and analyzed immediately after formation and the same sample after 24 h in reaction mixture, i.e., pure DADP.

Table 4. Influence of sulfuric acid concentration on the composition of the products (reaction temperature 20 °C, all peaks were used to determine the 100% area).

Molar Ratio n_c/n_a	Acid Concentration (%)	Peak Area (%)		
		DADP	TATP	Peak 10.2
0.25	31	0	92.4	7.3
	65	0	91.4	8.1
	96	0	91.2	7.9
0.5	31	0	92.7	6.6
	65	0	93.5	6.1
	96	0	94.0	5.2
1	31	0	93.2	6.0
	65	0	95.8	3.9
	96	42.8	54.6	1.5
2.5	31	0	95.4	4.4
	65	98.7	0	0
	96	99.5	0	0
5	31[a)]	–	–	–
	65	99.0	0	0
	96[b)]	–	–	–

[a)] Crystalline product did not form.
[b)] Mixture decomposed during mixing.

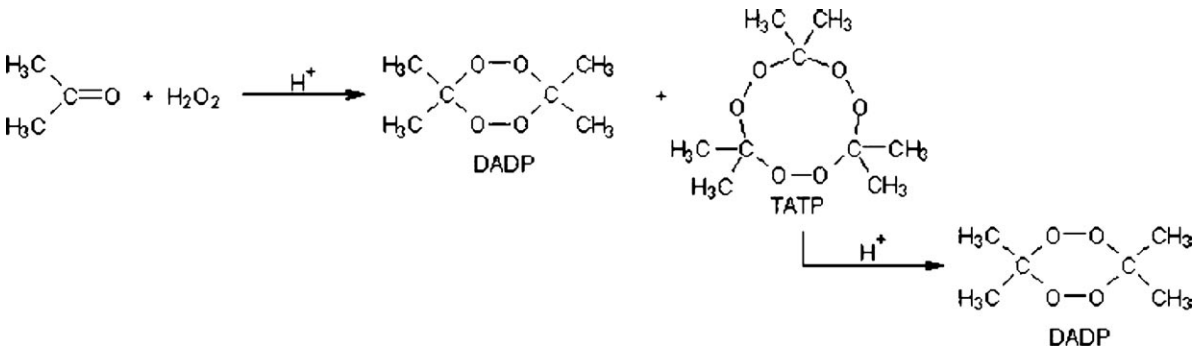
reaction mixture occurs. It has been observed that the reaction mixture first changes color locally to canary yellow just before the decomposition.

It was found that at higher molar ratio n_c/n_a DADP and TATP mostly form in the reaction mixture during the addition of the catalyst, but pure DADP was identified in the same sample after product drying. This means that TATP spontaneously transforms to DADP in a strongly acidic environment (Scheme 2). This phenomenon was observed in the case of sulfuric acid (65%), perchloric, hydrochloric, and nitric acid. Two chromatograms of the product separated at different times (prepared from perchloric acid at $n_c/n_a = 2.5$) are shown in Figure 2. The spontaneous transformation of TATP to DADP was already observed during aging of dry TATP prepared from perchloric and sulfuric acid at higher molar ratio n_c/n_a [23, 24] and transformation of TATP to DADP was catalyzed by p-toluenesulfonic acid in dichlormethane solution [11].

3.1.2 *Influence of Catalyst Concentration*

The concentration of the catalyst is an important parameter that has an influence on the composition of the product. This parameter was investigated in the case of sulfuric acid. Three concentrations were used for TATP preparation – 31, 65, and 96%. The results are summarized in Table 4. The DADP forms when higher concentration of catalyst at higher molar ratios n_c/n_a is used. This is clearly demonstrated in Table 4. The catalyst with concentration of 31 and 65%, both yield TATP as a main product while the catalyst with a concentration of 96% yields DADP at the same molar ratio of $n_c/n_a = 1$. The yields are, however, low.

The concentration and the amount of the catalyst is not reported in some older papers which can lead to some confusion. For example, Pastureau reported [13] that DADP forms by reaction of acetone, 2% hydrogen peroxide, and sulfuric acid of unspecified concentration. We have observed that pure DADP forms as a product when concentrated sulfuric acid is used and TATP with side



Scheme 2. Reaction of acetone and hydrogen peroxide in the presence of very high concentrations of strong inorganic acid showing TATP as an intermediate in the formation of DADP.

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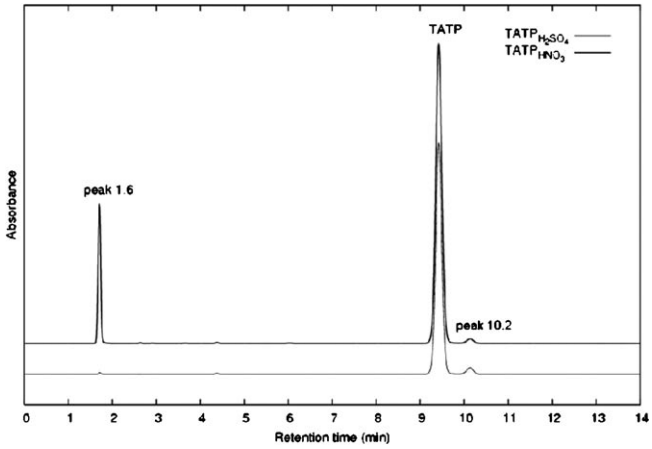


Figure 3. Chromatogram demonstrating the formation of peak of unknown substance with RT 1.6 min in samples formed from nitric acid. The sample from sulfuric acid is used as a reference; both samples were prepared with ($n_c/n_a = 1$).

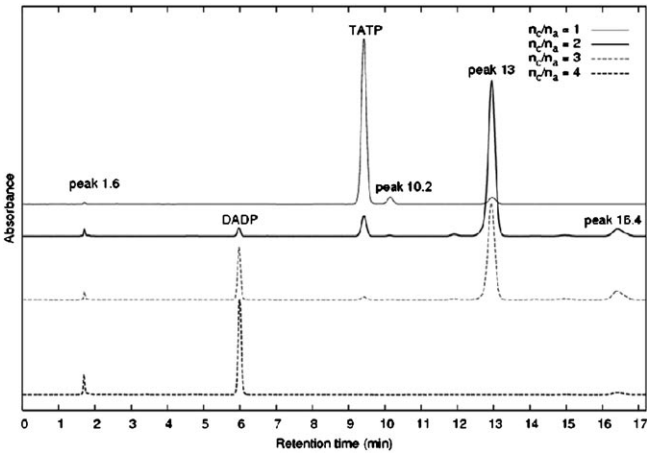


Figure 4. Chromatogram demonstrating the influence of high and very high concentrations of hydrochloric acid on product composition.

product (peak 10.2) when 31% sulfuric acid (in a very low yield) is used.

3.1.3 Nitric Acid Related Side Products

All types of samples showed a small peak at 1.6 min corresponding to the dead volume. The only exceptions were samples prepared using nitric acid, which showed this peak as well but with significantly larger area. A comparison of the TATP and the product from nitric acid is demonstrated in Figure 3 for $n_c/n_a = 1$. An increasing amount of nitric acid increased the amount of this unknown substance (Table 5). The size of the area observed in other samples (when other catalyst were used) was approximately constant (from 0.17 to 1.15%, average 0.53%), irrespective of the catalyst type or amount.

The unknown product having an RT of 1.6 min may be concentrated by sublimation of TATP at temperatures below the m.p. of TATP (used 85 – 90 °C). The sublimate, however, always contains some TATP and side product (peak 10.2). Recrystallization from acetone or dichloromethane results in pure TATP accompanied only by the side product (peak 10.2).

Table 5. Composition of the products prepared from nitric acid at different n_c/n_a (reaction temperature 20 °C, all peaks were used to determine the 100% area).

Molar Ratio n_c/n_a	Peak Area (%)			
	Peak 1.6	DADP	TATP	Peak 10.2
0.025	3.1	–	89.8	6.7
0.05	3.6	–	89.2	6.9
0.1	3.4	–	90.6	4.4
0.25	8.1	–	58.8	6.1
0.5	8.4	–	90.1	1.5
1	16.3	–	82.3	1.4
2.5	28.7	71.3	–	–

Table 6. The composition of the products prepared from hydrochloric acid at higher molar ratios n_c/n_a (reaction temperature 20 °C, all peaks were used to determine the 100% area).

Molar Ratio n_c/n_a	Peak Area (%)				
	DADP	TATP	Peak 10.2	Peak 13.0	Peak 16.4
1	–	90.1	4.3	5.2	–
1.5	–	15.8	0.8	79.4	3.3
2	1.9	7.5	0.4	83.8	5.5
2.5	14.0	2.0	–	74.2	9.2
3	16.9	1.4	–	70.4	9.8
3.5	39.7	–	–	42.5	16.2
4	83.9	–	–	–	6.1

3.1.4 Hydrochloric Acid Related Side Products

Two new side products were observed in the product when hydrochloric acid was used as the catalyst at higher molar ratios $n_c/n_a \geq 1$ (RT 13.0 and 16.4 min, respectively). These two side products were not observed with other catalysts.

The proportion of these side products and also of DADP increases with a higher molar ratio n_c/n_a from 1 to 3 when product RT 13.0 min becomes the dominant peak and the proportion of TATP and side product (peak 10.2) decreases significantly. DADP becomes a major product at even higher molar ratios $n_c/n_a = 3.5$ or 4. The higher molar ratio n_c/n_a, however, leads to a significant decrease in product yield. The typical chromatographs are shown in Figure 4 and the results are summarized in Table 6.

3.2 Influence of Temperature

The influence of the reaction temperature was studied for all the catalysts (HCl, H_2SO_4, HNO_3, $HClO_4$, CH_3SO_3H, $SnCl_4$, HCOOH, CH_3COOH). In this set of experiments, the molar ratio of catalyst to acetone n_c/n_a was 0.25. TATP forms as the main product, the side product (peak 10.2) as side

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Table 7. Influence of the reaction temperature on the composition of the TATP samples expressed as the ratio of chromatographic peak areas of TATP and the side product (peak 10.2) at $n_c/n_a =$ 0.25.

Catalyst	Temperature (°C)					
	Group A				Group B	
	0	10	20[a)]	30	−20	0
HCl	18.6	33.8	11.3	17.2	12.4	20.8
H_2SO_4 (31%)	18.8	21.2	17.1	16.0	13.6	14.0
HNO_3	18.5	30.0	12.5	15.1	13.1	26.8
$HClO_4$	18.6	32.9	29.1	37.0	14.2	26.7
CH_3SO_3H	17.4	18.4	27.4	14.6	14.2	12.2
$SnCl_4$	17.7	13.5	16.1	13.1	16.3	16.3
HCOOH	22.6	24.1	20.1	19.4	16.5	18.9
CH_3COOH	26.3	25.0	24.1	22.3	–[b)]	20.0

[a)] Average values from repeated experiments (see Table 8).
[b)] At this temperature TATP did not form within the period of the test.

product in all samples. The concentrations were in the interval from 91.0 to 96.7% TATP and from 2.9 to 8.1% side product (peak 10.2). Traces of other side products were occasionally detected in amounts below 0.2%. The temperature has been found to have negligible influence on the composition of the final product. This contradicts previously reported findings of Pacheco-Londono et al. and Pena Quevedo et al. [3, 21]. They reported formation of TATP/DADP mixture at temperatures above 10 °C. We did not find any traces of DADP under all temperature regimes. The results are summarized in Table 7.

The reaction temperature and acid strength of the catalyst, however, influence the rate of formation of TATP. This is especially apparent in the case of formic and acetic acids when for group A experiments TATP is formed in low yield (especially from acetic acid) and is not formed even after 7 days at −20 °C (from acetic acid) for group B. On the other hand, TATP starts to form already during the catalyst addition to reaction mixtures at higher temperatures (30 and 40 °C) in the case of strong inorganic acids.

3.3 Repeatability of the Results

The results of the dependence of the composition of prepared TATP on the temperature and catalyst type should be carefully considered before trying to create some correlations out of the measured data. The scatter in the results inherently includes batch to batch variation with unknown magnitude. For the purpose of evaluation of the level of this scatter five batches of TATP were prepared using each of the catalysts under exactly the same conditions (20 °C, $n_c/n_a = 0.25$). The results are summarized in Table 8.

3.4 Influence of the Order of Reagents' Mixing

The influence of the order of mixing (adding acetone to the mixture of hydrogen peroxide with catalyst or peroxide to the mixture of catalyst with acetone) was investigated for hydrochloric, sulfuric (31%), nitric, perchloric, and methanesulfonic acids of molar ratios n_c/n_a 0.5, 1, 2.5, and 5. The order of mixing did not influence the composition of the obtained product.

Table 8. Repeatability of composition of prepared samples expressed as ratio of peak areas of TATP/side product (peak 10.2) at 20 °C, $n_c/n_a = 0.25$.

Catalyst	Sample Number					Average	Standard Deviation
	1	2	3	4	5		
HCl	12.6	11.3	21.5	11.4	20.7	15.5	5.15
H_2SO_4	12.6	17.1	15.1	17.5	14.4	15.3	2.01
HNO_3	14.1	12.5	20.2	11.9	19.3	15.6	3.89
$HClO_4$	17.0	29.1	32.3	23.2	25.4	25.4	5.85
CH_3SO_3H	27.4	21.8	16.3	18.3	16.6	20.1	4.64
$SnCl_4$	16.1	21.8	16.3	20.3	18.2	18.5	2.49
HCOOH	20.1	18.2	18.0	18.0	18.9	18.6	0.90
CH_3COOH	24.1	18.7	18.4	18.2	21.7	20.2	2.60

4 Conclusion

The influence of reaction conditions (temperature, catalyst type, catalyst concentration – expressed as molar ratio of catalyst to acetone n_c/n_a) on the composition of the product formed from the reaction of acetone and hydrogen peroxide (30%) under acid catalysis was studied by HPLC. It was found that TATP forms as the main product and that it is always contaminated with small amounts of a side product when $n_c/n_a \leq 0.5$. This side product cannot be removed by sublimation or recrystalization.

The type of catalyst, its concentration, and reaction temperature do not have a significant influence on the composition of the product at $n_c/n_a \leq 0.5$; the formation of DADP was not observed in any sample. The order in which the reactants were mixed did not influence the composition of the product.

DADP forms as a side product and later as the main product at higher molar ratio $n_c/n_a \geq 1$ when strong inorganic acids are used as catalysts. The concentration of catalyst has an influence on the composition of the product. The proportion of DADP in the product increases with increase in the concentration of the catalyst. DADP was not observed in the formed product when acetic or formic acid was used, even at a molar ratio $n_c/n_a = 2.5$. The product in this case is only TATP with the above mentioned side product.

The transformation of TATP to DADP in the reaction mixture was observed in the case of strong inorganic acids at higher molar ratio $n_c/n_a \geq 1$. The initial product contained a high proportion of TATP that later transformed to DADP if left standing in the acidic reaction mixture.

Two new products were found when hydrochloric acid was used as catalyst at higher molar ratios $n_c/n_a \geq 1$. These side products were not observed with other catalyst. Another side product was observed in the case of use of

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nitric acid. Its proportion increased with increase in the molar ratio n_c/n_a. The structure of these side products was not determined.

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Acknowledgements

Our thanks are due to Prof. Anthony J. Bellamy for his useful comments on the manuscript. This work was created as a part of the project of Ministry of Education, Youth and Sports of the Czech Republic No. MSM-0021627501.

Symbols and Abbreviations

n_c/n_a	Molar ratio of acid catalyst to acetone
RT	Retention time

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